Rivyn

Proof-bound cross-chain invoice settlement

Abstract

Rivyn is a cross-chain escrow protocol that releases Creditcoin invoice funds only after Attestcoin proves a matching buyer approval on Ethereum Sepolia.

The problem

Cross-chain invoices create a verification gap: approval occurs on one network while settlement occurs on another. Screenshots, relayers, and manual claims do not provide enforceable evidence.

The protocol

Each escrow is bound to an invoice ID, token, amount, buyer, destination, source contract, expiry, and proof policy. The buyer approves the exact terms on Ethereum Sepolia. Rivyn records the commitment on Creditcoin and waits for proof.

Proof verification

Rivyn validates transaction inclusion, receipt success, event fields, source contract, chain key, Merkle siblings, continuity roots, expiry, and replay protection. Only a complete match can authorize release.

Fail-closed security

Missing, stale, malformed, expired, or mismatched proof is rejected. There is no trusted-backend fallback, manual override, or partial payout path.

USC and Attestcoin integration

USC contracts on Creditcoin custody invoice funds and execute the final release. Attestcoin supplies the cross-chain proof payload connecting the Sepolia approval to a verifiable source-chain history.

Demonstrated evidence

The public deployment includes a real Sepolia approval, Attestcoin proof data, Creditcoin escrow release, deployed contract addresses, source code, and API evidence.

Conclusion

Rivyn turns cross-chain settlement from a claim into a policy-checked, cryptographically evidenced action.